AUTOCOLLECTIVE AI × FETCH.AI

Project Summary, Technical Architecture & Live Demo Speech Guide

Fetch-A-Thon 3.0 | GCET National AI Agent Hackathon Edition · Comprehensive Teammate Briefing

Project: AutoCollective AI	**Event:** Fetch-A-Thon 3.0 (GCET)	**Team Handle:** @adityak1273
Live Portal: vercel.app deployed	**Agentverse Address:** agent1q09hp6x...	**Tracks:** Best Personal AI, Best Multi-Agent

■ 1. Project Summary & Core Problem Solved

The Problem: When an individual retail consumer buys a car in India, they negotiate in isolation. Dealerships exploit this informational asymmetry with inflated ex-showroom add-ons, non-negotiable dealer charges, and rigid retail margins. In contrast, corporate entities purchasing vehicle fleets routinely command **10% to 15% bulk discounts** directly from regional distributors.

The Solution: AutoCollective AI bridges this gap by transforming isolated retail car shoppers into autonomous bulk-purchasing clusters. Using Fetch.ai uAgents on Agentverse and ASI:One, the platform groups buyers by exact vehicle model, variant, and regional PIN code. Once a threshold (2+ buyers) is reached, an autonomous reverse-auction RFQ (Request For Quote) is triggered across competing regional dealership uAgents, unlocking wholesale corporate fleet pricing and saving individual buyers ■50,000 to ■3,50,000+ per vehicle.

Zero Mock Data Mandate: Unlike standard hackathon prototypes with hardcoded JSON prices, AutoCollective features a real-time web scraping proxy indexing live prices from CarWale, CarDekho, Spinny, and Cars24 across India. Luxury exotics (Porsche 911, Range Rover, Defender, Audi) are autonomously normalized into **Crores (■X.XX Cr)**, while mass-market cars (Tata Punch, Hyundai Creta) format in **Lakhs (■X.XX L)**.

■■ 2. Technical Architecture & End-to-End Workflow

Cloud Agent Gateway (Agentverse Cloud)	Fetch.ai uAgents SDK, Agent Chat Protocol (ACP v0.2.0), Almanac Ledger, ASI:One	Global autonomous public presence. Listens for user queries on ASI:One, verifies handshake via ChatAcknowledgement, scrapes live price benchmarks, and assigns pool hashes (#mumbai_tata_punch).
Multi-Agent Bureau (Negotiation Engine)	Python AsyncIO, uAgents Bureau, Pydantic v2 schemas, Dealer Margin Engine	5 concurrent agents: Personal Buyer Concierge (8000), Group Coordinator (8001), and 3 competing dealership uAgents (Metro Hyundai 8002, Apex Motors 8003, Star City 8004). Executes automated volume bidding.
Client Dashboard (Vercel & Docker)	HTML5 Neo-Brutalist layout, CSS3 Poster styling, Vanilla ES6+, Vercel Edge Server	High-engagement consumer web interface. Features 9-field buyer registration, dynamic vehicle image recognition, and live operational trading terminal feed streaming agent events.

End-to-End Execution Pipeline (6 Steps):

1. Demand & Identity Ingestion: User completes clean 9-field registration on the dashboard or types directly into ASI:One.

2. Multi-Source Web Query: Cloud agent or dashboard triggers DuckDuckGo proxy index across CarWale, CarDekho, Spinny & Cars24.

3. Currency Heuristic Normalization: Luxury cars are automatically categorized as Crores (■2.52 Cr for Porsche); mass-market cars stay in Lakhs.

4. Group Pool Hash Formation: Group Coordinator calculates deterministic pool key (e.g. #delhi_porsche_911) and updates group registry.

5. Dealership Reverse Auction: Once threshold ≥2 is met, RFQ broadcast causes Metro Hyundai, Apex Motors, and Star City Auto to submit volume-discounted bids.

6. Final Report & Deal Finalization: Coordinator selects the highest-discount winning bid and outputs final savings on the dashboard & ASI:One.

■ 3. 3-Minute Hackathon Demo Speech & Action Guide

Use this exact script for team presentations, pitch rounds, and screen recording demos. Follow the action cues to demonstrate the prototype side-by-side.

Minute 0:00 - 0:45 · The Hook & The Problem Statement

Speaking Script: *"Hello judges and innovators! When an everyday consumer walks into a car showroom in India, they stand completely isolated. Dealerships take advantage of this with opaque markups, inflated ex-showroom add-ons, and fixed pricing. But when corporate corporations purchase cars in bulk fleets, dealerships eagerly slash prices by 10 to 15 percent. Today, we are presenting AutoCollective AI—a decentralized multi-agent collective car buying network built on Fetch.ai uAgents that democratizes corporate fleet purchasing power for everyday retail consumers."*

■ Live Demo Action: ACTION: Display the live Vercel dashboard (https://fetch-a-thon-30-hackthon-guuor6t9b-adityak1273-8754.vercel.app/). Show the clean poster aesthetic, the top notification strip, and the 9-field buyer registration modal.

Minute 0:45 - 1:30 · Live Edge Prototype & Real-Time Intelligence

Speaking Script: *"Let us demonstrate how it works. On our live portal, a buyer configures their vehicle. Notice that we do NOT use fake static JSON or hardcoded lists. Watch what happens when I search for a Porsche 911 in Delhi versus a Tata Punch in Mumbai. The agent initiates real-time multi-source web queries across CarWale, CarDekho, Spinny, and Cars24. Notice the dynamic visual rendering and our currency heuristic: the Porsche benchmark is accurately tagged in Crores at ■2.52 Crore, while the Punch formats in Lakhs. The user can toggle between Solo Retail mode and Collective Group Pooling with a single click."*

■ Live Demo Action: ACTION: Type "Porsche 911" with city "Delhi" and click "Scrape & Query Web". Point out the live extracted snippet, ■2.52 Crore pricing, the sports car preview photo, and the live terminal feed scrolling in real-time.

Minute 1:30 - 2:15 · Multi-Agent Negotiation & Reverse Auction

Speaking Script: *"Now, here is where Fetch.ai uAgents come into play. When our Group Coordinator Agent detects two or more buyers targeting the same model and region, it generates a unique pool hash on the Almanac ledger and issues an autonomous Request For Quote. In our Bureau, three independent dealership uAgents—Metro Hyundai, Apex Motors, and Star City Auto—engage in an automated reverse auction. Because we have pooled collective demand, dealerships undercut each other, unlocking wholesale fleet discounts of 6 to 12 percent, plus free 3-year comprehensive warranties."*

■ Live Demo Action: ACTION: Switch to your terminal or show the Bureau execution log / dashboard terminal feed where Metro Hyundai, Apex Motors, and Star City Auto calculate margin discounts and submit bids.

Minute 2:15 - 3:00 · ASI:One Discovery & The Winning Close

Speaking Script: *"Finally, our system is natively integrated with ASI:One. Our agent @adityak1273 is live on the public Agentverse cloud. Anyone can search for our agent on asi1.ai and negotiate or join a pool in conversational English. In conclusion, AutoCollective AI bridges real-time web intelligence with Fetch.ai's decentralized agent economy to save Indian families lakhs of rupees on every car purchase. Thank you!"*

■ Live Demo Action: ACTION: Open ASI:One (https://asi1.ai/chat/d534e5c0-5443-4f6b-b318-536a9ab09921) and show the live conversation with @adityak1273.

■ 4. Jury Q&A; Defense Cheat Sheet

Q1: How is this different from existing portals like CarWale or CarDekho?

CarWale and CarDekho are static advertising lead-generation portals that sell user phone numbers to dealers as individual leads. AutoCollective AI is an autonomous negotiation engine that groups buyers before dealerships ever see them, forcing dealerships to bid competitively against each other.

Q2: Why use Fetch.ai uAgents instead of a standard Python backend?

Fetch.ai uAgents provide decentralized agent identities with cryptographic addresses on the Almanac ledger, autonomous peer-to-peer messaging via ACP, and native discoverability on ASI:One. Dealerships operate as sovereign software agents with private margin rules that never expose internal profit margins to competitors.

Q3: How do you prevent spam buyers in collective pools?

Every buyer profile contains 9 verified demographic and location attributes, and pool confirmation is anchored to cryptographic agent session state, preventing fake demand spoofing.

■ 5. Official Project Submission Links

Live Vercel Deployment	https://fetch-a-thon-30-hackthon-guuor6t9b-adityak1273-8754.vercel.app/
GitHub Repository	https://github.com/Adityak1273/FETCH-A-THON-3.0-HACKTHON.git
Agentverse Agent Profile	https://agentverse.ai/agents/details/agent1q09hp6xwguamvsrzd5xgqc8qxyhn3dfgf67pjpz6my68llydt8y9clcwqu9/profile
ASI:One Live Session	https://asi1.ai/chat/d534e5c0-5443-4f6b-b318-536a9ab09921